Image Text Enhancer: A Lightweight, Easy-to-Use, and Efficient C++ Library

Algorithm Engineering 2026 Project

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Abstract

TBC

Keywords

task parallelism, WebAssembly, OpenMP, image processing, C++

ACM Reference Format:

Daniel Motz, Leonard Teschner, and Mher Mnatsakanyan. 2026. Image Text Enhancer: A Lightweight, Easy-to-Use, and Efficient C++ Library: Algorithm Engineering 2026 Project. In *Proceedings of AEPRO 2026: Algorithm Engineering Projects (AEPRO 2026).* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

RQ Can a lightweight document enhancement pipeline be optimised to achieve strong multicore performance and practical WebAssembly deployment while preserving the enhancement behaviour of the baseline implementation?

RQ1. How much faster is the optimised pipeline than the baseline on native CPU execution?
RQ2. How well does the optimised pipeline scale under strong and weak scaling from 1 to 8 cores?
RQ3. What runtime overhead does the WebAssembly version introduce compared to native execution, and is that overhead still acceptable for practical use?

1.1 Background

The extraction of textual information from images — from scanned books to even receipts captured by smartphone cameras — have become an important part of everyday life [1, 8]. Any such extraction necessarily begins with the pre-processing of recorded images through techniques such as binarisation, noise removal, and geometric transformation, a process generally referred to as *document enhancement.* Numerous tools focused on this niche have been proposed. Although these approaches feature broad capabilities, they are often associated with substantial complexity on the user's end and entail considerable software dependencies. These dependencies restrict applicability, particularly in settings where binary size is constrained, such as mobile and web platforms. This limitation assumes added importance in a context where most digital media interactions now take place on smartphones and application development is increasingly oriented toward edge-computing models. Consequently, efficient and platform-independent implementations that exploit parallelisation and other optimisation techniques are of considerable importance for reducing processing time and enhancing user satisfaction.

Table 1: Comparison of related tools.

Tool	Lightweight	Licence	Commercial
ImageMagick	✗	Own	✓
Leptonica	✓	Own	✓
Unpaper	✗	Mixed	✗

1.2 Related Work

Image enhancement and document preprocessing are covered by a wide range of open-source tools and commercial software. The scope ranges from general purpose image processing and OCR libraries to specialised tools for post-processing scanned documents.

ImageMagick [4] is a suite for image processing that is available as a command line tool and as a library. It offers functions such as format conversion, resizing, filtering, and colour correction. However, ImageMagick is not specifically designed to enhance scanned or photographed images. Similar to OpenCV, the diverse features result in a large code base.

The C-based Leptonica library [3] is a general-purpose image processing and image analysis application. It is a dependency to both Tesseract and OpenCV. Although Leptonica is relatively lightweight compared to the other libraries, much like our approach, it primarily serves as a low-level library and does not offer ready-to-use, configurable processing pipelines.

A very similar tool to our approach is Unpaper [5], which focuses on post-processing scanned documents. It offers features such as deskewing, binarisation, and noise reduction that have been developed specifically for this purpose. The tool also offers a CLI variant, with which pipeline steps can be configured and disabled. Unpaper requires the library ffmpeg, which in the standard configuration consumes >100MiB of space.

1.3 Our Contribution

Unlike the existing tools and libraries described, our approach is designed to be minimalistic, extensible, and user-friendly. Instead of being a comprehensive image processing library or OCR engine, we focus on a lightweight, modular image enhancement pipeline, focused on printed and handwritten text, written entirely in C++ with only CImg as a dependency, which is itself is lightweight

has no further dependencies [7]. Image Text Enhancer (ITE) has a lightweight build footprint, requiring only three build dependencies: OpenMP, X11, and JPEG. It can be imported as a library and also as a CLI executable—with 1.7 MiB on Linux.

Compared to OpenCV and ImageMagick, our work is characterised by its deliberate focus on document enhancement and compactness. Unlike Leptonica, our pipeline is specialised in the post-processing of scanned documents and unlike Unpaper, our binary including dependencies is ≈50× smaller. In addition, our licensing model (MIT-0 and CeCILL-C) permits use in research and commercial applications. By combining minimal dependencies, a modular architecture, portability through C++ and OpenMP, and a commercially usable licence, our pipeline fills a gap between large-scale, general-purpose image processing libraries and highly specialised tools for document enhancement.

Furthermore, we systematically optimise and improve our baseline implementation and evaluate the performance gains through various experiments. In a proof of concept, we show that our approach can be executed on WASM platforms, with only minor performance drawbacks.

1.4 Outline

In section 2, we introduce the pipeline that we will implement and optimise. In section 3, we describe the optimisation techniques used. These are evaluated in section 4, where we compare the performance of the pipeline under different settings to the baseline implementation. Finally, we provide an outlook on possible future work 5.

2 Pipeline

The pipeline is designed to be modular, enabling users to activate, deactivate and configure each stage as required, both through the CLI and the library. Some stages can intentionally not be disabled. These exceptions are marked in algorithm 1 with an **asterisk**.

Algorithm 1 Image Text Enhancer Pipeline

(1) Grayscale Conversion*
(2) Deskew
(3) Contrast Enhancement
(4) Denoise: (Adaptive) Gaussian and Median Blur
(5) Binarisation*: Otsu, Sauvola, Bataineh
(6) Morphological Operations
(7) Colour passthrough

The user can choose between different algorithms for certain stages—as these have different effects on the result, users can select the algorithm best suited to their application. For instance, Bataineh's adaptive binarisation method with automatic window adjustment [2] is helpful when binarising historical, handwritten documents with uneven writing and lighting, whereas Otsu's method [6] is a good choice for binarising typewritten documents with uniform font and lighting. Standard configurations are provided for each step, enabling the pipeline to be used straight away. These configurations deliver good results for most use cases and are based on fixed recommendations from the authors. This means that the pipeline can be applied immediately. However, users can also customise the configurations to optimise the pipeline and improve their specific results.

We provide a command-line application that can be used independently of a program to access the pipeline. This enables the pipeline to be configured and used directly via the command line.

3 Optimisations

We hypothesise that the baseline implementation fails to scale linearly with problem size, although such scaling would be desirable for image processing workloads. We therefore concentrated on identifying low-hanging fruits at the level of high-level algorithmic design, for example by selecting more efficient algorithms for specific tasks, while adhering to best practices such as avoiding false sharing and writing code amenable to auto-vectorisation.

3.1 Parallelisation and Concurrency

To achieve better performance on multicore processors, the individual steps are parallelised using OpenMP. *parallel for collapse(3)* environments are mainly used for the pixel processing loops. However, some binarisation algorithms also use other parallelisation strategies such as the reduction clauses *reduction(min)* and *recuction(max)*. When implementing parallelisation, care is taken to ensure that no race conditions or false sharing occur. For example, thread-local buffers are used for contrast enhancement to avoid false sharing and locking. In the deskewing logic, detection and application of the correction are separated. The angle is detected once and then applied to the grayscale and color image to avoid unnecessary calculations.

3.2 Memory Management and Allocation

To prevent increased memory consumption and long runtimes, the image is not copied at each step. Instead, the steps work in-place on the input image. For this purpose, a reference is passed through the pipeline. This approach is continued in grayscale conversion by replacing the final deep copy task of the image with a pointer swap of the original image pointer with the new one. Memory optimisations were made in further steps. For example, in the adaptive Gaussian blur filter and the adaptive median filter, the buffer allocations are removed from the pixel processing loop to reduce the number of memory allocations. Sauvola's binarisation algorithm uses a lightweight fused vector builder instead of the *8-bit CImg<double>* image structure. Similarly, the core utilities return preallocated buffers instead of heavy objects.

3.3 Algorithmic Complexity Reduction

Several different algorithms are implemented in the pipeline. Some of these are optimised through algorithmic improvements to reduce complexity. One example is Bataineh binarisation, which reduces the number of operations per pixel from $O(w^2)$ to $O(w)$—where w is the window size—by using integral images to calculate the number of black and red pixels in the window instead of iterating over the entire window. In morphological operations, the complexity is reduced from $O(n^2)$ to $O(2n)$—where n is the number of pixels—by decomposing the 2D kernel operations into two 1D

kernel operations for the horizontal and vertical dimensions. In addition, the sliding window min/max algorithm is implemented with a monotonic queue to enable computation for arbitrary window sizes in amortized $O(1)$ time. In the deskewing logic, the "rotate and project" method is replaced by a “Radon transform” method. Instead of moving the pixels, the coordinates of the foreground are extracted and mathematically projected.

3.4 Loop level Optimisations

To reduce the runtime of pixel processing loops, logical optimisations are made to the main loops that iterate over the pixels. In contrast enhancement, for example, the linear stretching formula is calculated in advance and stored in a lookup table. As a result, the main loop performs an $O(1)$ query per loop iteration instead of floating-point calculations. In addition, the if/else branch in the pixel processing loop is replaced by std::min clamping. This allows the compiler to auto-vectorize the loop with SIMD. In Sauvola binarisation, a branchless lookup is also implemented to improve performance.

3.5 Architectial Refactoring

In some cases, architectural changes were also made to improve performance. For example, in the deskewing logic, detection and application of the correction were separated. The angle is first detected and then applied to the grayscale and color images to avoid unnecessary calculations. In the core utilities, pre-built buffers are returned instead of creating heavy objects. This improves performance.

4 Experiments

4.1 Setup

To evaluate the performance of the pipeline with the optimisations, various experiments are conducted and the results are compared with the baseline implementation. The experiments examine the runtime when processing images of different sizes with and without parallelisation.

The experiments assess three aspects: (1) single-threaded runtime versus image size, (2) eight-threaded runtime versus image size, and (3) strong-scaling behaviour with increasing core count for a large fixed image size.

All stages of the pipeline are enabled, with Bataineh binarisation and the adaptive Gaussian and adaptive median filters selected.

For our measurements, we use a machine with an AMD Ryzen 9950X3D 16-core processor (32 hyperthreads) running WSL2 with Ubuntu 24.04.1 LTS with 64 GB of RAM[1]. Each core has a base frequency of 4.3 GHz, with boost frequency of around 5.5 GHz. Care has been taken to limit the number of background processes, so as not to affect performance.

We selected a small but representative set of document images spanning both document type and resolution. The benchmark includes textual documents of various sizes and degrees of detail (see Table 2 for details). These choices aim to capture common document-enhancement scenarios while also testing scalability to very large inputs.

[1]It is clear that this comes with a performance penalty, however the results will still be conclusive. See this blog post which tested the penalty on very similar hardware.

4.2 Results

Figures 1 and 2 show the development of runtime with increasing image resolution for the baseline and optimised implementations. Unsurprisingly, the runtime t increases with the problem size s. We found, that the optimised implementation scales almost linear, surpassing the baseline. For $c_1, c_2 < 1.7 \cdot 10^{-5}$ we observe

$$t_{\text{baseline}} \approx c_1 \cdot s^{1.660}$$
$$t_{\text{optimised}} \approx c_2 \cdot s^{1.071}$$

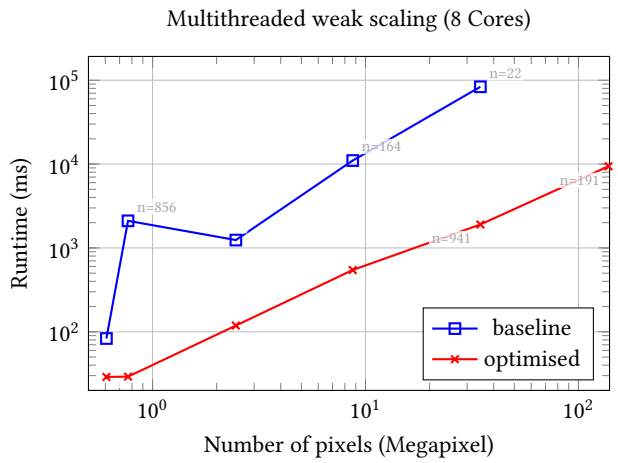

Figure 1: Runtime as a function of image size for the baseline and optimised implementations on 8 cores. Each plotted point corresponds to the mean runtime over repeated measurements for the respective input image. Labels are shown only for measurements whose sample count differs from the standard case of 1000 repetitions. For clarity, all but one documents with image size of $\approx 10^1$ Megapixels were omitted, as the results were almost identical.

The results of the individual phases show a significant reduction in runtime in the deskewing and Bataineh binarisation phases when comparing the baseline and optimised implementations. Table 3 shows the runtime of the individual phases for the KITA test image with a resolution of 3840x2254 on 8 cores. It can be seen that the optimisations lead to a massive reduction in phase runtime of over 98% for Bataineh binarisation in particular. The deskewing phase also benefits from the optimisations and shows a reduction in phase runtime of over 98%. In other phases, such as grayscale and the filter, a significant reduction in phase runtime can also be observed, while in the morphological ops phase, a slight increase in phase runtime can be seen. WHY?????. Overall, this leads to a reduction in the total runtime of over 95%. In absolute terms, the runtime improvement is mainly due to the optimisation of Bataineh binarisation, as this phase accounts for the largest share of the total runtime in the baseline implementation.

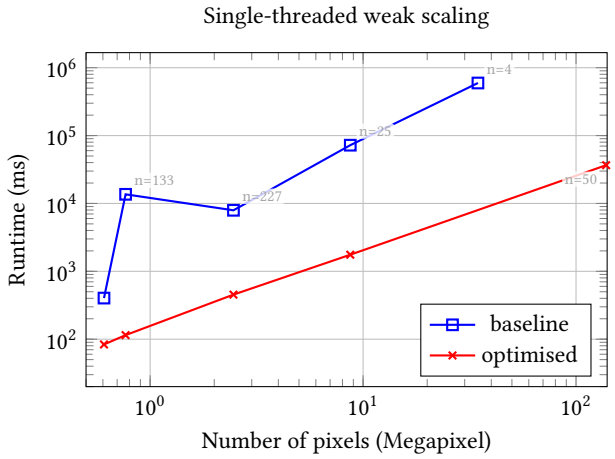

Figure 2

Table 2: Comparison of total runtimes (8 Cores)

Testbild	resolution	baseline (ms)	optimised (ms)	Speedup
Text Small 1K	1017x597	130,577	42,511	3,07×
Text 2K	2048x1202	1.770,138	188,975	9,37×
Magazine 2K	2516x3272	16.701,170	649,924	25,70×
Text 4K	3840x2254	16.328,084	781,634	20,89×
Text 8K	7680x4508	114.607,067	3.015,428	38,01×
Text 16K	15360x9017	-*	15.181,484	-*

*Could not be calculated due to massive runtime

Table 3: Comparison of phase runtimes (3840x2254, 8 Cores)

Phase	baseline (ms)	optimised (ms)	diff (%)
Init & Copy	65,458	63,672	-2,728%
Grayscale	30,950	17,753	-42,640%
Deskew	**159,405**	**3,062**	**-98,079%**
Filters (Sum)	483,312	380,467	-21,279%
Bataineh	**15.507,105**	**217,347**	**-98,598%**
Morphological Ops	66,663	93,05	+39,583%
Color Preservation	4,129	4,089	-0,969%
Total	**16.328,084**	**781,634**	**-95,213%**

Table 4 shows the detailed phase runtime for the different phases and images. It can be seen that the optimisations lead to a significant reduction in runtime in all phases, especially in Bataineh binarisation, which accounts for the largest share of the total runtime in the baseline implementation. For larger images such as the 8K text image, it can be seen that the image becomes too large for the system, as there is a sharp increase in the Init & Copy phase, which indicates a memory limitation.

5 Future Work

(1) Use heuristics for selecting algorithms, e.g. binarisation
(2) better deskew
(3) runtime experiments on WASM
(4) try to optimise for WASM (reduce the performance gap)
(5) What do we do to avoid false sharing?
(6) Look into blocking code

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Table 4: Detailed runtimes for certain pipeline stages

Phase	1K Small	2K Text	2K Mag.	4K Text	8K Text
Init & Copy	1,29	4,20	61,83	65,46	241,93
Grayscale	0,60	2,95	27,68	30,95	92,45
Deskew	24,22	59,89	28,90	159,41	518,54
Binarisation	71,00	1.555,57	15.999,83	15.507,11	111.495,99
Filter (Sum)	28,74	125,14	520,73	483,31	1.968,43
TOTAL	**130,58**	**1.770,14**	**16.701,17**	**16.328,08**	**114.607,07**